

Kyle M. Booker

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Experience

Reviewer, AI Training (Programming, Mathematics)

Vancouver, B.C.

Outlier AI

January 2024 — Present

- Run, debug, and fix contributor-submitted Python, C, C++, and JavaScript code across math, programming, and physics tasks; verify correctness, edge-case handling, and runtime behaviour before submissions enter the training pipeline.
- Audited 5,000+ submissions, identifying logical errors and bugs and authoring structured feedback that improved contributor consistency and downstream model accuracy.

Data Scientist

Vancouver, B.C.

IntelliFi — Residential Technology

June 2022 — March 2023

- Shipped an Azure Computer Vision OCR ingestion service (Python) processing 1,000+ monthly PDFs end-to-end, replacing manual data entry and cutting analyst workload by 70%.
- Built and deployed supervised forecasting models to Azure with REST inference endpoints integrated into production systems.
- Owned the reporting layer over the data pipeline, exposing KPIs to product, sales, and executive stakeholders via Power BI.

Data Analyst

Vancouver, B.C.

CMLS Financial — Residential Capital Markets

June 2021 — June 2022

- Built and operated an automated Python/SQL ETL pipeline processing 1 GB of customer data per hour, replacing manual workflows.
- Engineered a Selenium ingestion service (Python) scraping 30,000 real-estate listings per week into a PostgreSQL store for analytics.
- Delivered a production churn-prediction service (Python, pandas) at 80% recall, driving a 15% lift in retention via targeted campaigns.

Graduate Research Assistant

Waterloo, Ont.

University of Waterloo

September 2018 — April 2021

- Implemented H(div)-conforming discontinuous Galerkin methods for multiphase flow in C++ on the MFEM finite-element library — 2,000+ lines including custom discretizations, solvers, and numerical experiments.
- Ran large-scale simulations on Compute Canada HPC clusters with parallel jobs spanning many cores and multi-day runtimes; built Python and ParaView pipelines for post-processing, convergence analysis, and visualization.
- Authored an NSERC grant proposal, securing \$27,000 in thesis research funding.

Undergraduate Research Assistant

Kamloops, B.C.

Thompson Rivers University

May 2016 — August 2018

- Refactored 500 lines of legacy C code and parallelized batch simulations with Bash and Python, doubling data throughput.
- Implemented Python and C algorithms for applied math research in graph theory, matroid theory, and numerical analysis.

Skills

Programming Python, SQL, JavaScript, TypeScript, C, C++, C#, Java, Bash, HTML/CSS

Infrastructure & DevOps Docker, K3s, Linux, Git, CI/CD, Cloudflare, VPN, REST APIs

Data & Cloud PostgreSQL, MongoDB, Redis, Apache Spark, pandas, AWS, Azure, Power BI

Education

University of Waterloo

Waterloo, Ont.

Master's in Applied Mathematics — Thesis: H(div)-conforming DG methods for multiphase flow

September 2018 — April 2021

Thompson Rivers University

Kamloops, B.C.

B.S. in Computer Science and Mathematics — GPA: 4.21 / 4.33

September 2014 — April 2018

Publications

Edge-coloured graph homomorphisms, alternating paths and duality — JCMCC

August 2022

On accuracy of numerical solutions to BVPs on infinite domains with slow decay — MMNP

April 2019

Projects

Self-Hosted Server

Vancouver, B.C.

Home-Lab Infrastructure

April 2023 — Present

- Built and operate a Linux home lab running 100+ Docker containers on K3s and Portainer, serving 20+ active users at 99% uptime.
- Production-style ingress with Traefik reverse proxy, automated TLS, Cloudflare tunnels, and WireGuard VPN for secure remote access.
- Self-hosted PostgreSQL and Gitea backing internal services; automated backups, container updates, and incident recovery.